Code No: 124AG **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year II Semester Examinations, March - 2022
FORMAL LANGUAGES AND AUTOMATA THEORY
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Convert the moore machine to determine residue mod 3 into mealy machine.
b) Check whether the following two FSM's are equivalent. [7+8]

M1	0	1
→A	B	D
(B)	A	C
C	D	B
(D)	C	A

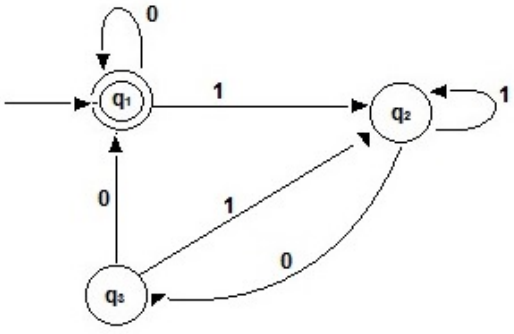
M2	0	1
→ P	R	R
Q	R	P
(R)	P	Q

2.a) Construct NFA with ε which accepts a language consisting the strings of any no. of 0's followed by any no. of 1's followed by any no. of 2's. And also convert into NFA without ε.
b) Convert the following NFA with ε to equivalent DFA. [8+7]

	0	1	ε
→A	Φ	A	B
B	B	Φ	C
Ⓒ	B	A	Φ

3.a) Construct the left linear grammar for the language (a+b)*aa(a+b)*
b) Prove that the following language $\{a^n b^n \mid n \geq 1\}$ is not a regular. [7+8]

4.a) Design a FA for the following Languages
i) (0*1*)*
ii) (a*bb*+bab)
b) Construct regular expression for the following finite automata: [7+8]



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5.a) Find the GNF equivalent to the following
$A \rightarrow BB \mid a$
$B \rightarrow AA \mid b$

b) Design a push down Automata for the language
$L=\{WCW^R \mid W \text{ is in } (0+1)^*\}$. [7+8]

6.a) Construct the CFG for the PDA
$M=(\{q_0,q_1\}, \{0,1\}, \{R,Z_0\}, \delta, q_0, Z_0, \Phi)$ and δ is given by
$\delta(q_0,1,Z_0)=(q_0,RZ_0)$
$\delta(q_0,1,R)=(q_0,RR)$
$\delta(q_0,0,R)=(q_1,R)$
$\delta(q_1,0, Z_0)=(q_0, Z_0)$
$\delta(q_0,\varepsilon, Z_0)=(q_0,\varepsilon)$
$\delta(q_1,1,R)=(q_1,\varepsilon)$

b) Eliminate Unit productions in the grammar
$S \rightarrow AB$, $A \rightarrow a$, $B \rightarrow C$, $B \rightarrow b$, $C \rightarrow D$, $D \rightarrow E$, $E \rightarrow a$. [8+7]

7.a) Design a TM to recognize the language $L = \{0^n1^n0^n \mid n \geq 1\}$.

b) Write a short note on Church's hypothesis. [7+8]

8.a) What are undecidable problems? Explain with example.

b) Check whether the following post correspondence problem has a solution or not. [7+8]

I	List A	List B
1	111	11
2	1001	01
3	1110	1001

---ooOoo---

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